BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18

1- ہر سوال کے چار ممکنہ جوابات [illegible in source scan] 2- جوابات کی شیٹ (A، B، C، D) کے لیے نیلے یا کالے بال پوائنٹ کا استعمال کریں۔ 3- جواب میں ایک سے زیادہ دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. Which of the following ions has largest heat of hydration

(A) Ba^{+2}
(B) K^{+1}
(C) Li^{+1}
(D) Be^{+2}

Q2. Molecular formula of silica is:

(A) SiO_4
(B) SiO_3
(C) SiO_2
(D) Na_2SiO_3

Q3. Which oxide sodium metal predominantly forms in oxygen?

(A) Na_2O
(B) Na_2O_2
(C) Na_2O_3
(D) NaO_2

Q4. What is the product when chlorine gas is passed over element silicon in powdered state on heated it produce colorless liquid having formula?

(A) $SiCl_2$
(B) $SiCl_4$
(C) Si_2Cl_3
(D) SiCl

Q5. A metallic oxide when added to water would most likely form a(n):

(A) Base
(B) Acid
(C) Salt
(D) Basic anhydride

Q6. Which one of the following forms the most acidic oxide:

(A) Al
(B) Si
(C) Fe
(D) P

Q7. Which one of following electronic subshells the lanthanides have in the process of filling?

(A) 4f
(B) 5f
(C) 5d
(D) 6d

Q8. Which of the following is not transition element?

(A) Zn
(B) Cr
(C) Mn
(D) Ni

Q9. Which of the following is not ferromagnetic substance?

(A) Iron
(B) Cobalt
(C) Nickel
(D) Barium

Q10. Complexes exists in various coordination numbers, choose the coordination number which is less common.

(A) 2
(B) 4
(C) 5
(D) 6

Q11. Esters are represented by the general formula.

(A) ROP
(B) BOOR
(C) RCOOR
(D) RCOOH

Q12. Octane number is associated with.

(A) Gasoline
(B) Kerosine oil
(C) Diesel oil
(D) Lubricating oil

Q13. A nucleophile is:

(A) Lewis acid
(B) Bronsted acid
(C) Bronsted base
(D) Lewis base

Q14. Reduction of acetaldehyde with H_2/Ni gives:

(A) Ethanol
(B) Ethanoic acid
(C) Ethane
(D) Ethylene

Q15. Acetic acid undergoes reduction with $LiAlH_4$ to give:

(A) Ethanal
(B) Ethane
(C) Ethyne
(D) Ethanol

Q16. Two glucose units combine by glycoside bond, the product known is:

(A) Sucrose
(B) Maltose
(C) Lactose
(D) Cellulose

Q17. All of the following are micronutrients except:

(A) Iron
(B) Chlorine
(C) Copper
(D) Potassium

Q18. In purification of water the coagulant used is:

(A) $NiSO_4$
(B) $BaSO_4$
(C) $CuSO_4$
(D) Potash alum

SECTION – B

Maximum Marks: 40

Note: Q-II: Answer any TEN parts. Each part carries FOUR marks.

1. Write down any four characteristics of group _ IIA elements.
2. Why fluorine is stronger oxidizing agent than chlorine?
3. Why the transition elements show the catalytic behavior?
4. How do transition elements differ from "s" and "p" block elements?
5. Why tertiary alcohol cannot be oxidized?
6. How bromine water test in alkene differ from bromine water test for phenol?
7. What is Lucas test? How Lucas test is used to distinguish alcohols?
8. Why phenol is more acidic than alcohol?
9. What are chemical entities that compose the nucleic acids?
10. Why vegetable oil is sensitive to rancidity?
11. Differentiate between reactive and non-reactive adhesives.
12. How nail polish remover, removes the nail polish?
13. Write any four principles of green chemistry.
14. Explain the nuclear spin flipping briefly.

SECTION – C

Maximum Marks: 27

> Note: Note: Attempt any THREE questions. All questions carry equal marks. Draw neat and labeled diagrams where necessary.

QIII. Q-III

(a) Explain halide ions as reducing agents and discuss their trend in reducing strength.

(b) Why the bond enthalpy is important in the thermal stability of hydrogen halides?

QIV. Q-IV

(a) Explain the bond enthalpy in halogens and bond enthalpies in halogen halides.

(b) Analyze the halide ions as reducing agents and trends in reducing strengths of halide ions.

QV. Q-V

(a) Write down the electronic structure of 3d-elements.

(b) Explain why the electronic configuration for chromium and copper differ from those assigned using the Aufbau principles.

QVI. Q-VI

(a) What are phenols? Give methods of preparation of phenols.

(b) Discuss the acidic behaviour of phenol.